



COMSATS University Islamabad, Lahore Campus

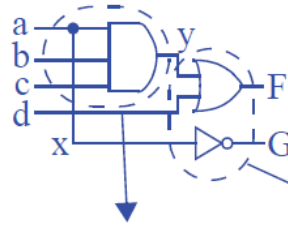
Department of Computer Engineering

Quiz 4 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A
Date:	18 th December, 2025	Maximum Marks:	10		
Student's Name:			Registration Number:		

Question 1: [CLO2-PLO3-C5] [10 Marks]

Design and implement the following function using any number of 3-input 2-output CLBs using switch matrix.



$$(a,b,c,d) = abc + d$$

$$(a) = a'$$

Inputs			Outputs		Inputs			Outputs	
a	b	c	x	y	y	d	x	F	G
0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	1	0	0
0	1	0	0	0	0	1	0	1	1
0	1	1	0	0	0	1	1	1	0
1	0	0	1	0	1	0	0	1	1
1	0	1	1	0	1	0	1	1	0
1	1	0	1	0	1	1	0	1	1
1	1	1	1	1	1	1	1	1	0

